

Fraud Detection Analysis

Advanced portfolio project

Measure rare fraud events, class imbalance, transaction concentration, and a transparent risk baseline.

project_id: fraud_detection | template_id: operations | source_rows: 48 | prepared_rows: 48 | generated_at: 2026-08-11T06:45:07.205083+00:00

Executive summary

Fraud Detection Analysis analyzed 48 rows across 6 fields. The reusable operations BI template presents source-backed KPIs, drivers, and detail tables. Total amount is \$20.2k with a transparent synthetic validation fixture or uploaded source.

Headline metrics

Rows analyzed

48

Total amount

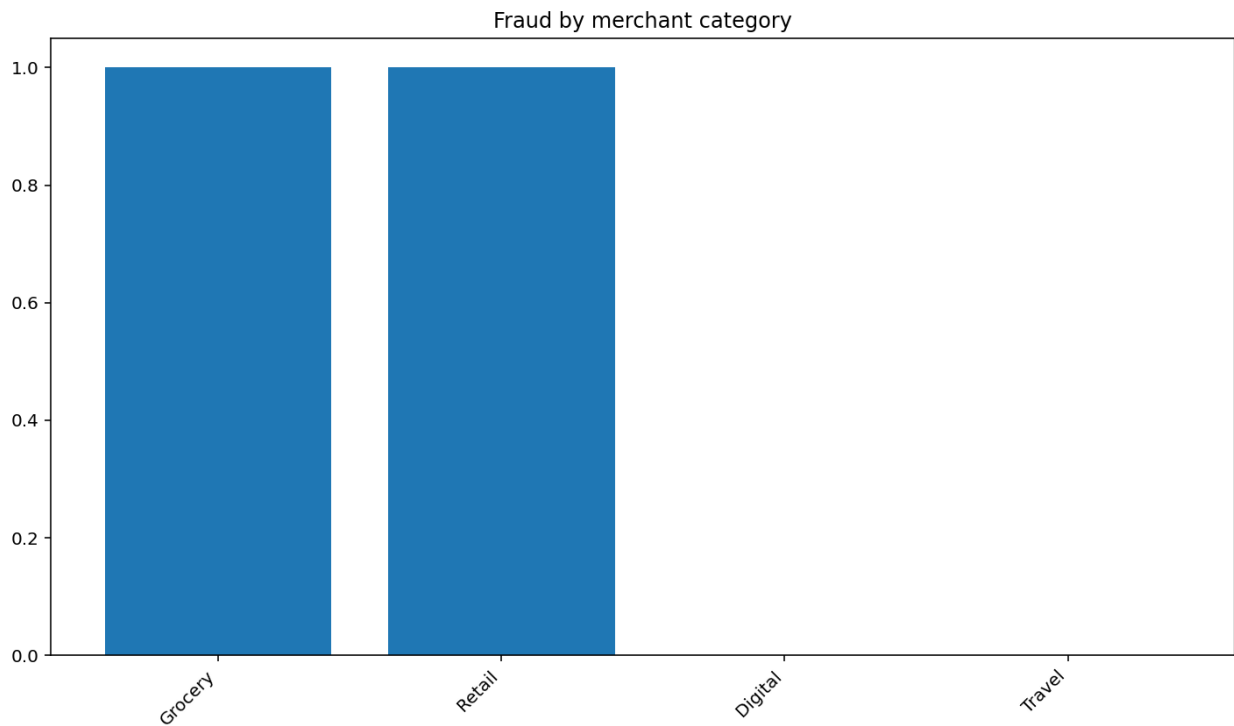
\$20.2k

Average amount

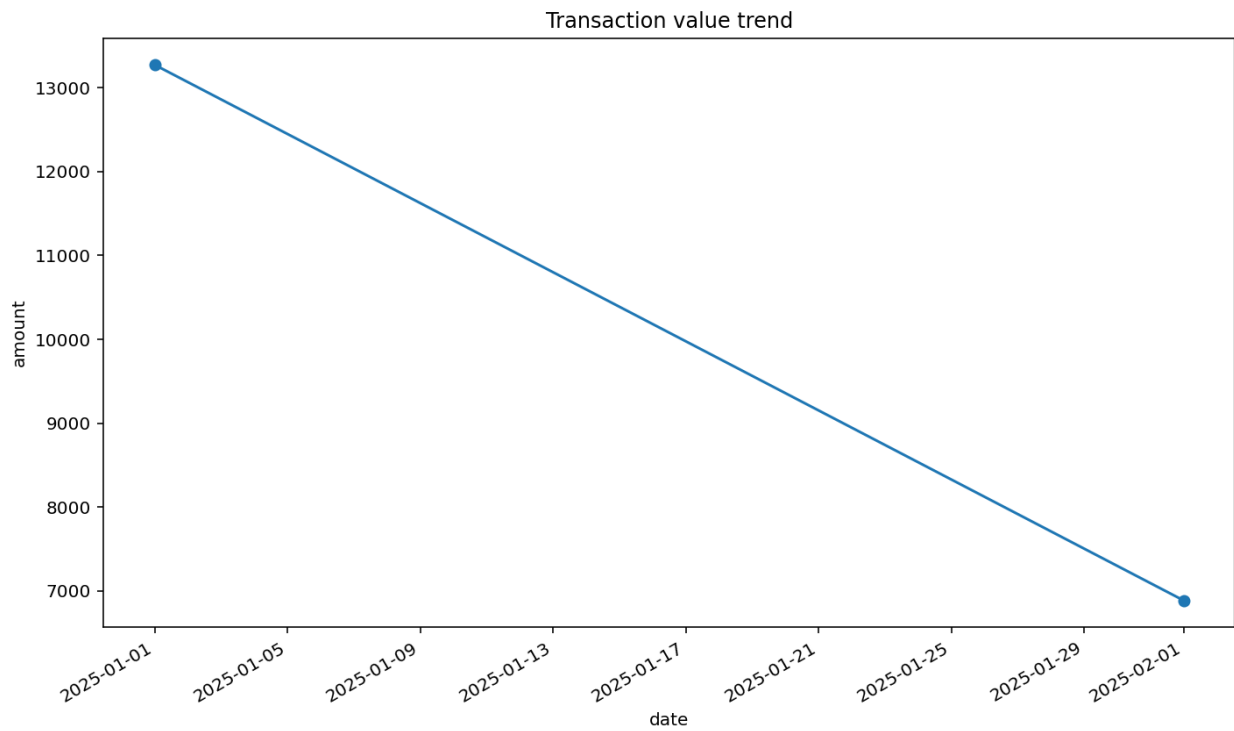
\$419.97

Drivers and trends

Fraud by merchant category



Transaction value trend



Detail by merchant_category

merchant_category	count	sum	mean
Grocery	12	1.0	0.08333333333333333
Retail	12	1.0	0.08333333333333333

merchant_category	count	sum	mean
Digital	12	0.0	0.0
Travel	12	0.0	0.0

Detail by date

date	count	sum	mean
2025-01-21T00:00:00	1	2829.2	2829.2
2025-02-17T00:00:00	1	754.6	754.6
2025-02-02T00:00:00	1	633.92	633.92
2025-01-25T00:00:00	1	617.35	617.35
2025-02-12T00:00:00	1	572.28	572.28
2025-01-24T00:00:00	1	571.39	571.39
2025-01-20T00:00:00	1	568.26	568.26
2025-02-06T00:00:00	1	563.99	563.99
2025-01-30T00:00:00	1	563.22	563.22
2025-01-14T00:00:00	1	561.55	561.55
2025-02-08T00:00:00	1	560.56	560.56
2025-01-15T00:00:00	1	553.14	553.14

Key findings

is_fraud is 95.8% zeros.; amount is nearly unique per row (100.0%).; date is nearly unique per row (100.0%).; transaction_id is nearly unique per row (100.0%).; amount is strongly right skewed (skew=4.51).; is_fraud is strongly right skewed (skew=4.74).; Log transform materially reduces skewness for amount.

Caveats and assumptions

Fixtures are synthetic validation data. Uploaded datasets remain the source of truth; field aliases are recorded in the build metadata.